HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2025)

Student No.: 20 Name: _____ Marks: / 32

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The polar coordinates of the points X, Y and Z are $(1, 20^\circ)$, $(2, 80^\circ)$ and (r, θ) respectively, where $20^\circ < \theta < 80^\circ$. If $\triangle XYZ$ is an equilateral triangle, find r.

   A. $\sqrt{3}$

   B. $\sqrt{5}$

   C. $\sqrt{7}$

   D. $\sqrt{10}$

(DSE-2025 PII Q24)

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2. $3E000000000000_{16} =$

A. $2^{16}+2^{15}+2^{14}+2^{13}+2^{12}$.

B. $2^{17}+2^{16}+2^{15}+2^{14}+2^{13}$.

C. $2^{52}+2^{51}+2^{50}+2^{49}+2^{48}$.

D. $2^{53}+2^{52}+2^{51}+2^{50}+2^{49}$.

(DSE-2025 PII Q31)

Answers written in the margins will not be marked.

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3. The L.C.M. of p^2-4q^2, p^3-8q^3 and $(p+2q)(p^2-4q^2)$ is

A. $p-2q$.

B. p^2-4q^2.

C. $(p+2q)^2(p^3-8q^3)$.

D. $(p+2q)(p^2-4q^2)(p^3-8q^3)$.

(DSE-2025 PII Q32)

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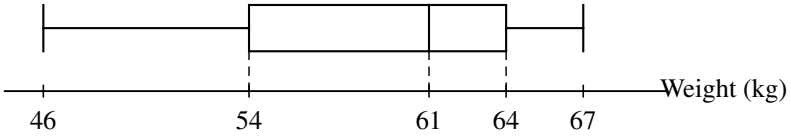
4. The stem-and-leaf diagram below shows the distribution of the weights (in kg) of some athletes before training.

Stem (tens)	Leaf (units)
4	5 6 8
5	w w 6 7
6	3 4 5 6 9 9
7	0 2
8	7

The difference of the range and the inter-quartile range of the above distribution is 25 kg.

(a) Write down the range of the above distribution. Hence, find w.

(b) The box-and-whisker diagram below shows the distribution of the weights (in kg) of the athletes after the training.



(i) Find the change in the upper quartile of the distribution due to the training.

(ii) Is the distribution of the weights of the athletes after the training less dispersed than that before the training? Explain your answer.

(7 marks) (DSE-2025 PI Q12)

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5. The straight line L cuts the positive x-axis and the positive y-axis at the points M and N respectively. Let P be a moving point in the rectangular coordinate plane such that $MP = NP$. Denote the locus of P by Γ. It is given that the equation of Γ is $3x - 2y - 30 = 0$.

(a) Describe the geometric relationship between Γ and MN.

(b) Find the equation of L.

(c) Suppose that Γ cuts L and the x-axis at the points Q and R respectively. Find the area of the quadrilateral $ONQR$, where O is the origin.

(7 marks) (DSE-2025 PI Q13)

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6. The equation of the straight line L is $4x-3y+83=0$. The equation of the circle C is $x^2+y^2+ax-2y+b=0$, where a and b are positive constants. It is given that C passes through the point $(-10,9)$ and L is a tangent to C.

(a) Prove that the equation $25x^2+(9a+640)x+(90a+4924)=0$ has equal roots. Hence, find a and b.

(b) The coordinates of the point P are $(-20,1)$. Let Q be a point lying below L such that PQ is tangent to C and $PQ=25$. Denote the centre of C by I.

(i) Find $\cos\angle IPQ$.

(ii) Find, in surd form, the distance between I and Q.

(iii) Let R be a point lying on L such that C is the inscribed circle of $\triangle PQR$. Denote the radius of the circumcircle of $\triangle PQR$ by r. Someone claims that $r>PQ$. Is the claim correct? Explain your answer.

(12 marks) (DSE-2025 PI Q19)

I can finish _____ questions

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